

that genetic engineering is at a very important stage. Before this, we could at best clone a human, and that too not very well. However, technology has been improving all the time, and now we're on the verge of understanding how DNA



Alana Saarinen has 99% of her DNA from her parents, and 1% from a mitochondrial donor. Read the full story on the BBC site: <http://bbc.in/1qiuRGa>

works, and have started building tools that will let us edit the very coded information that makes us human. It's now that the people of the world will take the call to decide what's OK, and what isn't.

As of 2015, we've seen a few dozen genetically modified humans born and grown up, and all from a two-year experiment in the US aimed at helping women conceive.

Some women have a problem with mitochondria in their eggs which causes them to be unable to conceive. What some scientists did was take mitochondria from a healthy woman's eggs and insert it into the other's, and then fertilise the eggs with the father's sperm, and the resulting fertilised eggs were the first ever humans with DNA from three different humans!

Not only is this a method to help women conceive, it's also a way to prevent those with markers for mitochondrial diseases from passing this on to their kids. Some of the most common mitochondrial diseases we know of are autism spectrum disorders, diabetes and thyroid problems.

Although the studies were banned in about two years in the US, that was enough for between 20 to 30 kids to be born using the method, and technically, have three "parents" in terms of DNA.

Since then, UK has made it legal after much public consultation and

speaking to experts, and is currently in the process of working out licensing structures in order to prevent misuse. Again, we should specify that this isn't designer babies in the making, because there's no controlling of traits other than preventing mitochondrial diseases – it's not like they're asking for only blonde-haired, blue-eyed kids! However, we do have to wonder what's next?

FOOD

Genetically modified food has been around for some time now, and like everything else that's controversial, there are two very obstinate and argumentative sides, and neither is willing to relent.

The naturalists will have you believe that we need to live naturally, and we need to go on vegan diets, or paleo diets, or whatever is the fad of the month, and they certainly don't want you to have anything that's even sprayed with pesticides... genetically engineered plants are just a clear no-no for them. Now there are plenty of arguments about why such people distrust anything remotely scientific, and our

medicine, and not be selective and run to get cancer treatment when death comes knocking for them or their loved ones.

The other camp is no better, to be honest, and they're often thought to be the big pharma and Monsanto apologists. Often accused of being on the payroll of big corporations, these people are very vocal about opposing the naturalists. Of course, because they're a more educated and scientifically-literate lot, they seem to make more sense. However, arrogance and turning a blind eye to all questioning is not wise either. Now we're not suggesting, for example, that there's any truth to the whole anti-vaxxer propaganda, or suggesting that vaccines be discontinued or anything like that. There's enough and more information to show that vaccines are safe. However, when it comes to genetics, it's still a new field, and thus it's not like we know for sure that there's no long term benefits. Claiming that is, in fact, very unscientific – yet that's exactly what many claim.

There is, however, one truth that we cannot hide from; when it comes down to it, the world will soon not be capable of feeding itself using natural methods. Thus, purely for the future of the human race, genetic engineering is needed. For plants



Greater than 90 per cent of the corn growing in the US is genetically modified to withstand pests and diseases

own guess is that they're also quite likely to be the anti-vaccine types, but we're sure that the world cannot sustain an all natural way of living – not when 7 billion humans bustling for space is the most unnatural thing around. If you come across such people, ask them to live truly natural, and shun all technology and

and animals, however. Until we know more, or have mastered genetic engineering for lower level life forms, we have no business meddling with human genetics.

What we're doing is akin to taking the best paper airplane designers in the world, and asking them to design fighter jets... it's just a little too soon for that...